

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Winter Examination – 2022**

**Course: B. Tech.**

**Branch: Chemical**

**Semester: V**

**Subject Code & Name: BTCHO 504B Pollution Control in Process Industries**

**Max Marks: 60**

**Date: 08/01/2023**

**Duration: 3 Hr.**

**Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

|                                                                                                                                                                                                                       | (Level/CO) | Marks     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------|
| <b>Q. 1 Solve Any Two of the following.</b>                                                                                                                                                                           |            | <b>12</b> |
| A) Explain in detail Carbon Cycle with proper schematic diagram.                                                                                                                                                      | CO1        | 6         |
| B) Explain in detail Nitrogen Cycle with proper schematic diagram.                                                                                                                                                    | CO1        | 6         |
| C) What are the consequences of population growth on the environment?<br>Example in detail with respect to forest, cropland, pasture and rangeland,<br>loss of biodiversity, fisheries and fresh water with examples. | CO1        | 6         |
| <b>Q.2 Solve Any Two of the following.</b>                                                                                                                                                                            |            | <b>12</b> |
| A) Explain in detail the following properties of air pollutants;<br>i) Particulate matter<br>ii) Carbon Monoxide<br>iii) Hydrocarbons<br>iv) Nitrogen oxides                                                          | CO2        | 6         |
| B) Explain in detail the following for particulates and gases;<br>i) Wet precipitation<br>ii) Dry deposition                                                                                                          | CO2        | 6         |
| C) Explain in brief effect of air pollution on human health.                                                                                                                                                          | CO2        | 6         |
| <b>Q. 3 Solve Any Two of the following.</b>                                                                                                                                                                           |            | <b>12</b> |
| A) Write in brief the cyclone separators with proper diagram.                                                                                                                                                         | CO3        | 6         |
| B) Explain in detail with proper diagram the fabric filters.                                                                                                                                                          | CO3        | 6         |
| C) What do you mean by electrostatic precipitation? Explain in detail about<br>electrostatic precipitators with neat diagram.                                                                                         | CO3        | 6         |
| <b>Q.4 Solve Any Two of the following.</b>                                                                                                                                                                            |            | <b>12</b> |
| A) What do you mean by desulphurization of flue gases? Explain in detail with<br>proper diagram Westvaco process.                                                                                                     | CO4        | 6         |

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|-----------|---------------------------------------------------------------------------|-----|----------|
| <b>B)</b> | Explain control of nitrogen oxides and control of hydrocarbons in detail. | CO4 | <b>6</b> |
| <b>C)</b> | Explain in detail types of water pollutants and their effects.            | CO4 | <b>6</b> |

**Q. 5 Solve Any Two of the following.** **12**

- |           |                                                                              |     |          |
|-----------|------------------------------------------------------------------------------|-----|----------|
| <b>A)</b> | Explain in detail determination of organic matters by the following methods; | CO5 | <b>6</b> |
|           | i) Dissolved Oxygen (DO)                                                     |     |          |
|           | ii) Biochemical Oxygen Demand (BOD)                                          |     |          |
|           | iii) Chemical Oxygen Demand (COD)                                            |     |          |
| <b>B)</b> | Explain in detail with proper diagram;                                       | CO5 | <b>6</b> |
|           | i) Zone settling                                                             |     |          |
|           | ii) Trickling filters                                                        |     |          |
| <b>C)</b> | Explain in detail sludge treatment and disposal of wastewater.               | CO5 | <b>6</b> |

**\*\*\* End \*\*\***